

# MicroPulse — Quadrant View Guide

## Reading the Gabbett 2017 internal vs external load framework

MicroPulse Performance Intelligence Platform

April 2026

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## 1. What this page tells you

The Quadrant view (Monitoring → Quadrant) is a single-screen visual that splits your squad into four groups by comparing **what the body actually did** (external load — distance, sprints, accelerations, captured by Catapult) against **what it cost the player to do it** (internal load — perceived effort, captured by daily RPE).

It is built on the **Gabbett 2017** framework, which is the most cited reference in modern football periodisation. The core insight is simple: the relationship between external work and internal cost predicts injury risk and fitness adaptation better than either signal in isolation. A player whose internal cost is rising while their external work is staying flat is on the edge of overreaching, even though their session sheet looks normal.

This page gives you that read for the entire squad in one glance. It updates immediately whenever new GPS or RPE data lands.

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## 2. The four quadrants

The chart is divided by team-median crosshairs. Where each player lands relative to those crosshairs places them in one of four quadrants:

### 2.1 Top-right — Injury risk (red background)

**High external load, high internal cost.** These players are doing a lot AND finding it hard. This is the classic pre-injury profile: the body is signalling it cannot keep absorbing what's being asked of it. Sustained presence in this quadrant is the strongest week-to-week predictor of soft-tissue injury in the published literature.

**What to do:** prescribe recovery. Cut session intensity for the next 2–3 days, pull them out of the highest-load drills, prioritise sleep + nutrition messaging.

## 2.2 Top-left — Decoupled (amber background)

**Low external load, high internal cost.** Players who are doing relatively little but feeling it heavily. This is the early-fatigue warning. Often appears 1–2 days before a player would otherwise show up in the red quadrant.

Common drivers: poor sleep, illness incoming, accumulated fatigue from previous weeks, life stress, or a previously protected player whose readiness hasn't yet caught up to a returning training load.

**What to do:** check the player's wellness check-in for context. Drop training intensity rather than volume — let the legs keep moving but take the spike out.

## 2.3 Bottom-right — Peak fitness (green background)

**High external load, low internal cost.** The player is doing the most work AND finding it easy. This is what every periodisation plan is trying to produce — high adaptation with low cost. Very often these are your most prepared starters during the in-season.

**What to do:** progress them. They have headroom. This is also the quadrant where you can legitimately push someone toward higher demand without worrying about overreaching.

## 2.4 Bottom-left — Under-stimulated (gray background)

**Low external load, low internal cost.** The player isn't doing much and isn't feeling much either. Common during recovery weeks, deload phases, or for fringe squad members not getting enough drill time.

**What to do:** depends on context. During a planned deload this is exactly where the squad should be. Outside a deload it is a sign to increase exposure — either give them more match minutes or add an extra session, otherwise they will not be ready when called on.

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# 3. Reading the page

## 3.1 Time window selector — top right

Three buttons: **7d** / **14d** / **28d**. They control which window of data the chart averages over.

- **7d** — most reactive. Shows the *current* state of the squad. Use for daily / weekly decisions.
- **14d** — smooths week-over-week noise. Best for periodisation discussions in the staff meeting.
- **28d** — chronic load view. Best for longer-term decisions like return-to-play readiness, contract / loan discussions, monitoring across an international break.

The page defaults to 7d and remembers your last choice.

## 3.2 Count cards — top of page

Four colored count cards above the chart, one per quadrant: **Injury risk, Decoupled, Peak fitness, Under-stimulated**. They tell you instantly how many players are in each group.

A healthy squad pattern in mid-season is roughly: 2 risk / 3 decoupled / 6 peak / 7 under-stimulated for an 18-player squad. Significantly more in risk + decoupled together (>30% of squad) means the team-wide load is too high — flag it for the head coach, not just for individual players.

## 3.3 The chart itself

- **X axis** — average external load (m/day) in your selected window
- **Y axis** — average internal cost (sRPE AU/day) in your selected window
- **Crosshairs** — the team median on each axis (so the four quadrants always have ~25% of your squad each on a balanced team)
- **Each dot** — one player. Hover to see their name and exact values

### Dot size — ACWR overlay

The size of each dot encodes that player's **Acute:Chronic Workload Ratio (ACWR)**:

- **Small dot** — ACWR below 1.3. Healthy chronic adaptation.
- **Medium dot** — ACWR between 1.3 and 1.5. Watch list.
- **Large dot** — ACWR above 1.5. The Gabbett “danger zone” — significantly elevated injury risk regardless of which quadrant they sit in.

A large dot in the top-right quadrant is the worst combination on the page. Treat it as a hard stop signal.

## 3.4 The “How to read this” helper text

Below the chart there is a short legend explaining each quadrant. It mirrors what's in this guide, in case you have a new staff member who needs the orientation. Everything you need to interpret the page is on the page.

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# 4. Workflow — how to use this in a normal week

## 4.1 Monday morning planning meeting

Open the Quadrant view at 14d. Scan for:

- Any large dots (high ACWR) — these get individual modification regardless of quadrant
- The shape of the cluster — is it shifted up-right (team-wide overreach) or down-left (team-wide deload)?

- Players who have moved quadrant since last week (you'll need to remember from previous Monday or compare to a screenshot)

Set the week's training intensity bias from this view. If half the squad is in risk + decoupled, the week should be lighter than planned even if the schedule says otherwise.

## 4.2 Friday — match selection input

Switch to 7d before squad selection. Players with elevated ACWR (large dots) in the risk quadrant are hard nos for starting unless a starter is unavailable. Decoupled-quadrant players can start but should be prepared for an early sub.

## 4.3 Mid-week — single-day check

A quick 7d glance after every training session lets you catch fast-moving players. Someone who jumped from peak fitness to decoupled in one session is signalling acute fatigue from that session — adjust the next day's plan accordingly.

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# 5. Data requirements — when this page is useful

The Quadrant view needs **both** GPS data **and** RPE check-ins to plot a player. If a player is missing one of the two, they won't appear on the chart.

- **No GPS data** — usually because the player was unavailable that day (injury, illness, missed session). Check Monitoring → Squad for context.
- **No RPE data** — the player didn't fill in their daily check-in. The system imputes a value where confident, but if the imputation also failed (e.g. player was off the squad), they drop off the chart.
- **No data at all** — there will be a “Add GPS / RPE data →” link below the chart pointing you to where to wire that up.

For new clubs: you need at least 7 days of paired GPS + RPE before this page becomes useful. With 28 days of data the ACWR overlay becomes meaningful.

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# 6. How this fits with the rest of MicroPulse

The Quadrant view gives you the **shape** of the squad. Other pages give you the depth on each player:

- A red-quadrant player → check **Decel Intelligence** to see *which dimension* of load is most stressful.
- A decoupled player → check the **Today / Decision Summary** to see what their wellness check-in says, and whether they have an active injury or illness.

- A team-wide overload pattern → check **Indoor Load** if the recent sessions were indoor — sometimes the spike is FMP-driven rather than GPS-driven.

The Quadrant view is the strongest *single screen* in the platform for explaining squad load to a head coach who doesn't want to read five different pages. Take a screenshot, drop it in the staff group chat, talk through it.

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## 7. Troubleshooting

### “All my dots are clustered in one corner”

This usually means the team-median crosshairs are skewed by one or two outlier players (often a returning-from-injury player whose load is artificially low). The quadrants are *relative to the team*, so an extreme outlier compresses everyone else.

Switch to 14d or 28d to smooth the effect, or temporarily exclude the outlier from the team for analysis. The page does not support manual exclusion yet — that's on the roadmap.

### “I can't see a player I know was at training”

They are missing either GPS or RPE for the window. Open Monitoring → Squad and check their data presence directly. Most common cause: the player skipped their RPE check-in, or wore a Catapult pod assigned to someone else.

### “ACWR dots all look the same size”

You don't have 28 days of data yet. ACWR needs a chronic 28-day baseline to compute. The dot-size encoding becomes visible once you have ~21+ days of paired data per player.

### “Window selector doesn't change anything”

Check the timestamp under the title — it should update whenever you change window. If it doesn't, your browser may have a stale render. Hit refresh (Cmd-R or Ctrl-R).

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*Last updated: 28 April 2026 · MicroPulse Performance Intelligence Platform · Methodology: Gabbett 2017 (External vs Internal Load); Hulin 2014 (ACWR)*